

TRENTOS-M on Microsemi PolarFire SoC with CycloneTCP

Page Content

- [Effort Estimation](#)
 - [Polar Fire](#)
 - [CycloneTCP](#)
 - [To be done:](#)
- [PolarFire SoC FPGA](#)
 - [General information](#)
 - [Development Boards](#)
 - [Porting](#)
 - [Renode](#)
- [CycloneTCP](#)

Sub-Pages

- [2020-05-12 Meeting notes - ORYX CycloneTCP Telco](#)

This page was created to provide information about the estimated effort for porting our TRENTOS-M to the PolarFire chip and using CycloneTCP instead of picoTCP.

Requested by	Unknown User (saskeg01) , Julius Rüdiger
Execution by	Thomas Böhm , Unknown User (sebeck01)

- [Effort Estimation](#)
 - [Polar Fire](#)
 - [CycloneTCP](#)
 - [To be done:](#)
- [PolarFire SoC FPGA](#)
 - [General information](#)
 - [Development Boards](#)
 - [Porting](#)
 - [Renode](#)
- [CycloneTCP](#)

Effort Estimation

[Thomas Böhm](#) / [Unknown User \(sebeck01\)](#) could you please do a breakdown of the tasks for PolarFire and CycloneTCP. 20 Apr 2020

Polar Fire

Task	Best Case Effort	Worst Case Effort
Setup development enviroment (TFTP Server) and get Linux running on the board (should work out of the box)	1 day	Should work out of the box, if it doesn't there would be big issues we can't plan for ATM.
Create a sel4 device tree. ▪ reuse an existing Linux one (should come with the board) ▪ might still need some modifications to run on the board (nodes we have to define by hand) ▪ worse case we have to create it all by ourselves	1 week	2-3 weeks (we might have other hardware related blockers)
Adapt hardware generation script (mapping memory, interrupt numbers) • depends on the board	a few days	1-2 weeks (if debugging / trial and error is required)
Implement platform specific kernel function (putchar)	1 day	-

Integrate userland drivers into the port (clock, gpio, i2c, serial, timer, ...)	5 days	2-3 weeks
Support multicore on RISC-V (multi-kernel as alternative)	3-4 weeks (needs some research time to create a correct effort estimate).	???

CycloneTCP

Task	Best Case Effort	Worst Case Effort
Include NIC Driver into TRENTOS	CycloneTCP already has drivers for the MAC /PHY included 1 day	We need to port linux drivers (with the assumption that the PolarFire SOC driver has a compatible license). - 5 days We need to implement it ourselves - ???
Implemented a ChanMux driver (CycloneTCP includes the driver in the NW stack)	5 days	5 days
Extend our Network Stack wrapper to work with CycloneTCP (a clean abstraction layer that supports different implementations underneath). A quick hack solution would be quicker.	2 weeks	2 weeks

To be done:

- implement the mutex / semaphore types/function stubs on our platform
 - 1 semaphore used to limit connection number
 - 4-5 mutexes used (need to find a way to use CAMKES sanely)
- wants a timer / function returning monotonically increasing time in ms
- nic interface is interrupt driven (RX/TX interrupts)
 - RPC interface
- stack configuration is done through a verbose header file (configurable library)

PolarFire SoC FPGA

General information

- <https://www.microsemi.com/product-directory/soc-fpgas/5498-polarfire-soc-fpga>

Development Boards

- Aries M100PFS:
 - <http://linuxgizmos.com/linux-powered-module-charges-up-the-risc-v-polarfire-soc/>
 - <https://www.aries-embedded.com/polarfire-soc-fpga-microsemi-m100pfs-som-mpfs025t-pcie-serdes>
- Microsemi PolarFire SoC Icicle Kit
 - <https://www.cnx-software.com/2019/12/11/risc-v-based-polarfire-soc-fpga-and-devkit/>
 - SOC has a 4x RISC-V subsystem and an FPGA
 - comes with 2 x Gigabit Ethernet also
 - available for shipping by Sep 15th for backers spending \$500 via <https://www.crowdsupply.com/microchip/polarfire-soc-icicle-kit/updates>
- inofficial Expansion Board for the HiFive Unleashed Board via Crowd Supply [sold out]: <https://www.crowdsupply.com/microsemi/hifive-unleashed-expansion-board>

Porting

- <https://docs.sel4.systems/projects/sel4/kernel-contribution.html>
- <https://docs.sel4.systems/projects/sel4/porting>
- ~~Multicore not supported on Risc Platform. Multi-Kernel might be a workaround.~~
 - Gernot: Bei uns läuft RISC-V multicore schon seit einiger Zeit (auf dem HiFive). D.h. für den PolarFire bräute es nur eine Plattform-Port, das ist keine große Sache, und DörnerWorks hat das ja anscheinend schon (aber warten wohl noch auf die Freigabe). Ich sollte Anfang nächste Woche eine Aufwandsabschätzung bekommen, für den Fall, das der DW-Pot nicht zur Verfügung steht.
- <https://github.com/polarfire-soc>
 - PolarFire SoC bare metal / yocto / buildroot
 - what hardware platforms are supported?
 - MACHINES:
 - mpfs: HiFive Unleashed board with the Microsemi's HiFive Unleashed Expansion kit.
 - lc-mpfs: Microchip's PolarFire FPGA using the Sifive U540 processor on a single board."

Renode

- PolarFire SoC emulation:
- <https://www.microsemi.com/product-directory/soc-fpgas/5498-polarfire-soc-fpga#getting-started>
- https://www.microsemi.com/document-portal/doc_download/1244824-polarfire-soc-renode-webinar-1

CycloneTCP

- CycloneTCP user manual - ONLY FOR INTERNAL USE
- The Network interface handle contains a pointer to the nic driver.



ORYX_CycloneTCP_User_Manual_v1.9.6.pdf